

A singleton spectral boundary is not a semicircle

Disproof of Conjecture 00000007157

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October 8, 2026

Statement and interpretation. Both supplied language versions assert that the spectral boundary of a nonnormal complex matrix is a semicircle whose radius is its numerical radius. We use “spectral boundary” in its standard literal sense, the topological boundary of the spectrum in $\mathbb{C}$. The counterexample below rules out every center and orientation of the proposed semicircle. It does not refute an alternative statement giving a semicircular enclosure or an inequality: such a statement is different from saying that the boundary *is* a semicircle. No enclosure formula or additional hypothesis is specified in the source.

Theorem 1. *The nonnormal matrix*

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has spectral boundary $\partial\sigma(J) = \{0\}$ and strictly positive numerical radius. Consequently its spectral boundary is not a semicircle of its numerical radius.

Proof. First,

$$JJ^* = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad J^*J = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

so J is nonnormal. For any complex z ,

$$\det(zI - J) = \det \begin{pmatrix} z & -1 \\ 0 & z \end{pmatrix} = z^2.$$

Thus $zI - J$ is invertible exactly when $z \neq 0$, and $\sigma(J) = \{0\}$. A singleton in $\mathbb{C}$ is closed and has empty interior, hence its topological boundary is itself.

For a unit vector $v = (v_1, v_2) \in \mathbb{C}^2$ with the Euclidean norm,

$$v^*Jv = \overline{v_1}v_2, \quad |v_1|^2 + |v_2|^2 = 1.$$

Therefore the numerical radius, by its actual quadratic-form definition, is

$$w(J) = \sup_{|v_1|^2 + |v_2|^2 = 1} |\overline{v_1}v_2|.$$

The inequality $(|v_1| - |v_2|)^2 \geq 0$ gives $|\overline{v_1}v_2| = |v_1||v_2| \leq 1/2$. Conversely, the rational vector $v = (3/5, 4/5)$ is a unit vector and gives $|\overline{v_1}v_2| = 12/25$. Hence

$$0 < \frac{12}{25} \leq w(J) \leq \frac{1}{2}.$$

These bounds, rather than an assumed numerical-radius formula, suffice.

Let $r = w(J) > 0$. Any closed semicircle of radius r can be expressed as

$$S(c, u, r) = \{c + u\zeta : |\zeta| = r, \operatorname{Im} \zeta \geq 0\}, \quad c \in \mathbb{C}, \quad |u| = 1.$$

Both $c + ur$ and $c - ur$ belong to this semicircle, and they are distinct because $u \neq 0$ and $r > 0$. They cannot both belong to $\{0\}$. Thus $\partial\sigma(J) \neq S(c, u, w(J))$ for every center and orientation. The same elementary cardinality obstruction also applies if “semicircle” excludes its endpoints: a positive-radius open semicircular arc still contains more than one point. $\square$

Formal verification. The Lean project represents J as a genuine 2×2 complex matrix, and checks nonnormality using conjugate transpose and matrix multiplication. Its spectrum is Mathlib’s algebraic spectrum: invertibility of $zI - J$ is reduced to its determinant, establishing `spectrum_singleton`. The standard topological frontier yields `boundary_singleton`. The formal numerical radius is the real supremum of the moduli of v^*Jv over all Euclidean unit vectors, with the Euclidean unit condition written explicitly in coordinates. The proof establishes nonemptiness, boundedness, and the bounds $12/25 \leq w(J) \leq 1/2$. The final theorem `boundary_not_semicircle` quantifies over arbitrary complex centers and unit complex orientations. It uses the actual positive numerical radius, not a scalar substitute or an assumed spectral certificate. This directly negates the conjecture’s universal boundary assertion.